

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Hydrazine hydrate, 55% (Hydrazine, 35%)

**CAS No** 10217-52-4

**Product Code** 296810000; 296810050; 296815000; 296810025; 296811000

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
No hazards identified

### **Health hazards**

|                                    |              |
|------------------------------------|--------------|
| Acute Oral Toxicity                | Category 3   |
| Acute Dermal Toxicity              | Category 3   |
| Acute Inhalation Toxicity - Vapors | Category 2   |
| Skin Corrosion/Irritation          | Category 1 B |
| Serious Eye Damage/Eye Irritation  | Category 1   |
| Skin Sensitization                 | Category 1   |
| Carcinogenicity                    | Category 1B  |

### **Environmental hazards**

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

### **Label Elements**



Skull and Crossbones



Health Hazard



Corrosion



Environment

**Signal Word**

**Danger**

**Hazard Statements**

H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H330 - Fatal if inhaled  
H350 - May cause cancer  
H410 - Very toxic to aquatic life with long lasting effects  
H301 + H311 - Toxic if swallowed or in contact with skin

**Precautionary Statements**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves  
P284 - Wear respiratory protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component           | CAS No     | Weight % |
|---------------------|------------|----------|
| Hydrazine           | 302-01-2   | -        |
| Hydrazine (hydrate) | 10217-52-4 | 50 - 60  |
| Water               | 7732-18-5  | 40 - 50  |

## Section 4 - First Aid Measures

**Inhalation**

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                                                                                                                               |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Nitrogen oxides (NO<sub>x</sub>), Ammonia, Hydrogen.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

#### Conditions for Safe Storage, Including any Incompatibilities

Store under an inert atmosphere. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

#### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component | Australia                                     | New Zealand WEL                                           | ACGIH TLV             | The United Kingdom                                                                                                                        | Germany |
|-----------|-----------------------------------------------|-----------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Hydrazine | TWA: 0.01 ppm<br>TWA: 0.013 mg/m <sup>3</sup> | TWA: 0.0002 ppm<br>TWA: 0.00026 mg/m <sup>3</sup><br>Skin | TWA: 0.01 ppm<br>Skin | STEL: 0.03 ppm 15 min<br>STEL: 0.039 mg/m <sup>3</sup> 15 min<br>TWA: 0.01 ppm 8 hr<br>TWA: 0.013 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin | Haut    |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Exposure Controls

##### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particulates filter conforming to EN 143 Inorganic gases and vapours filter Type B Grey conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**Particle filtering: EN149:2001 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

|                                                |                                                 |                                          |
|------------------------------------------------|-------------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                                       |                                          |
| <b>Physical State</b>                          | Liquid                                          |                                          |
| <b>Odor</b>                                    | Ammonia-like                                    |                                          |
| <b>Odor Threshold</b>                          | No data available                               |                                          |
| <b>pH</b>                                      | 12                                              | 350 g/l aq.sol                           |
| <b>Melting Point/Range</b>                     | -65 °C / -85 °F                                 |                                          |
| <b>Softening Point</b>                         | No data available                               |                                          |
| <b>Boiling Point/Range</b>                     | 109.4 °C / 228.9 °F                             |                                          |
| <b>Flash Point</b>                             | > 100 °C / > 212 °F                             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available                               |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 9.3 Vol%<br><b>Upper</b> 83.4 Vol% |                                          |
| <b>Vapor Pressure</b>                          | 15 mbar @ 20 °C                                 |                                          |
| <b>Vapor Density</b>                           | 1.1 @ 15 °C                                     | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.023                                           |                                          |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                   |
| <b>Water Solubility</b>                        | Miscible                                        |                                          |
| <b>Solubility in other solvents</b>            | No information available                        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                          |
| Hydrazine                                      | -0.16                                           |                                          |
| <b>Autoignition Temperature</b>                | 310 °C / 590 °F                                 |                                          |
| <b>Decomposition Temperature</b>               | No data available                               |                                          |
| <b>Viscosity</b>                               | 1.26 mPa.s at 20 °C                             |                                          |

**Explosive Properties** No information available  
**Oxidizing Properties** No information available

#### Other information

**Molecular Formula** H4 N2 . x H2 O  
**Molecular Weight** 32.04

## Section 10 - Stability and Reactivity

**Reactivity** No

**Stability** Do not allow evaporation to dryness. Air sensitive.

**Conditions to Avoid** Exposure to air, Incompatible products.

**Incompatible Materials** Acids, Bases, Finely powdered metals, Halogens, nitrogen oxides (NOx), Organic materials, Peroxides, Lead, Metals, copper, Butyl rubber.

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Ammonia. Hydrogen.

**Hazardous Polymerization** Hazardous polymerization does not occur.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 2 |

#### Toxicology data for the components

| Component | LD50 Oral               | LD50 Dermal                | LC50 Inhalation                              |
|-----------|-------------------------|----------------------------|----------------------------------------------|
| Hydrazine | LD50 = 60 mg/kg ( Rat ) | LD50 = 91 mg/kg ( Rabbit ) | 570 ppm ( Rat ) 4 h<br>0.75 mg/L ( Rat ) 4 h |
| Water     | -                       | -                          | -                                            |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | Category 1        |

**Sensitization** No information available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | Australia | New Zealand | New South | Western | IARC | EU | UK | Germany |
|-----------|-----------|-------------|-----------|---------|------|----|----|---------|
|-----------|-----------|-------------|-----------|---------|------|----|----|---------|

|           |  |                      |              |                  |          |              |  |        |
|-----------|--|----------------------|--------------|------------------|----------|--------------|--|--------|
|           |  |                      | <b>Wales</b> | <b>Australia</b> |          |              |  |        |
| Hydrazine |  | Suspected carcinogen |              |                  | Group 2A | Carc Cat. 1B |  | Cat. 2 |

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## Section 12 - Ecological Information

**Ecotoxicity effects** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component           | Freshwater Fish                                                                                                                                                                                                                                                                                        | Water Flea | Freshwater Algae                                                                                                                                                                                 | Microtox                                                                     |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Hydrazine           | LC50: 0.28 - 1.34 mg/L, 96h static (Poecilia reticulata)<br>LC50: 1.81 - 2.79 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1.17 mg/L, 96h (Lepomis macrochirus)<br>LC50: 0.54 - 1.31 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.7 - 1.3 mg/L, 96h flow-through (Lepomis macrochirus) |            | EC50: = 0.006 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.02 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.071 mg/L, 72h (Pseudokirchneriella subcapitata) | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min |
| Hydrazine (hydrate) |                                                                                                                                                                                                                                                                                                        |            |                                                                                                                                                                                                  | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min |

### Persistence and Degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.  
**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Hydrazine | -0.16   | No data available             |

**Mobility** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                                                        |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with high pH-value must be neutralized before discharge. Do not let this chemical enter the environment. |

## Section 14 - Transport Information

### IMDG/IMO

|                             |                             |
|-----------------------------|-----------------------------|
| <b>UN-No</b>                | UN3293                      |
| <b>Proper Shipping Name</b> | HYDRAZINE, AQUEOUS SOLUTION |
| <b>Hazard Class</b>         | 6.1                         |
| <b>Packing Group</b>        | III                         |

### ADG

|                             |                             |
|-----------------------------|-----------------------------|
| <b>UN-No</b>                | UN3293                      |
| <b>Proper Shipping Name</b> | HYDRAZINE, AQUEOUS SOLUTION |
| <b>Hazard Class</b>         | 6.1                         |
| <b>Packing Group</b>        | III                         |

| Component      | Hazchem Code |
|----------------|--------------|
| Hydrazine      | 2X           |
| 302-01-2 ( - ) | 2WE          |

### IATA

|                             |                             |
|-----------------------------|-----------------------------|
| <b>UN-No</b>                | UN3293                      |
| <b>Proper Shipping Name</b> | HYDRAZINE, AQUEOUS SOLUTION |
| <b>Hazard Class</b>         | 6.1                         |
| <b>Packing Group</b>        | III                         |

|                               |                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Environmental hazards</b>  | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
| <b>Special Precautions</b>    | No special precautions required                                                                          |
| <b>Additional information</b> | None known                                                                                               |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations      **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**



Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component            | Standard for the Uniform Scheduling of Medicines and Poisons |
|----------------------|--------------------------------------------------------------|
| Hydrazine - 302-01-2 | Schedule 6 listed - present                                  |

#### Australian Industrial Chemicals Introduction Scheme (AICIS)

| Component            | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|----------------------|-------------------------------------------------------------|------------------------|
| Hydrazine - 302-01-2 | Present                                                     | -                      |
| Water - 7732-18-5    | Present                                                     | -                      |

#### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

| Component            | Australia | New South Wales | Western Australia | New Zealand          |
|----------------------|-----------|-----------------|-------------------|----------------------|
| Hydrazine - 302-01-2 |           |                 |                   | Suspected carcinogen |

#### International Inventories

| Component           | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|---------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Hydrazine           | X    | X     | 206-114-9 | -      | X    | X   | -    | X     | X    | X    | X     | KE-19981 |
| Hydrazine (hydrate) | -    | X     | -         | -      | -    | -   | -    | -     | X    | X    | X     | -        |
| Water               | X    | X     | 231-791-2 | -      | X    | X   | -    | X     | X    |      | X     | KE-35400 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component           | CAS No     | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------|------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Hydrazine           | 302-01-2   | Listed         | Not applicable                             | 0.5 tonne                                                                                 | 2 tonne                                                                                  |
| Hydrazine (hydrate) | 10217-52-4 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Water               | 7732-18-5  | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

#### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrazine | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - 206-114-9 - Carcinogenic, Article 57a                                           |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

#### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Revision Date** 17-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

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**End of Safety Data Sheet**